12TH CHEMISTRY · CHAPTER 28 · FULL CHAPTER MCQS

Chromatography

90 topic-wise MCQs from the complete chapter with answers and reasons

28.1 INTRODUCTION

1. Which technique is suitable for separating a complex mixture such as leaf pigments?

(a) chromatography (b) filtration
(c) simple distillation (d) decantation

Answer: (a) chromatography — Complex mixtures need advanced techniques.

2. Chromatography separates components on the basis of their:

(a) different masses only (b) different colours
(c) different affinities for two phases (d) different smells

Answer: (c) different affinities for two phases — A stationary and a mobile phase.

3. Chromatography allows identification, purification and:

(a) synthesis of substances (b) quantification of substances
(c) combustion of substances (d) polymerisation

Answer: (b) quantification of substances — With high precision.

4. Adsorption is a:

(a) type of distillation (b) bulk phenomenon
(c) nuclear change (d) surface phenomenon

Answer: (d) surface phenomenon — Particles accumulate only on the outer surface.

5. Activated charcoal is used in filters because it:

(a) dissolves impurities (b) adsorbs gases and impurities
(c) reacts with water (d) evaporates quickly

Answer: (b) adsorbs gases and impurities — On its surface.

6. Which one is NOT one of the three important types of chromatography in the chapter?

(a) ion chromatography (b) paper chromatography
(c) thin layer chromatography (d) gas chromatography

Answer: (a) ion chromatography — PC, TLC and GC are discussed.

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7. A pigment found in leaves and separated by chromatography is:

(a) keratin (b) haemoglobin
(c) chlorophyll a (d) insulin

Answer: (c) chlorophyll a — Others: carotene, xanthophyll, chlorophyll b, pheophytin.

8. The phase which moves and carries the components is the:

(a) solvent front (b) stationary phase
(c) baseline (d) mobile phase

Answer: (d) mobile phase — A solvent or a carrier gas.

9. An advantage of chromatography is that it uses:

(a) a very large quantity (b) a very small quantity of the mixture
(c) high pressure only (d) no solvent or gas

Answer: (b) a very small quantity of the mixture — Stated in the book.

10. Simple mixtures of one or two components can be separated by:

(a) mass spectrometry
(b) gas chromatography only
(c) filtration, distillation or solvent extraction
(d) X-rays

Answer: (c) filtration, distillation or solvent extraction — Simple techniques.

28.2 PAPER CHROMATOGRAPHY

11. The separating principle of paper chromatography is:

(a) difference in boiling point
(b) magnetism
(c) difference in density
(d) relative solubility of the components

Answer: (d) relative solubility of the components — Partition between two phases.

12. The stationary phase in paper chromatography is:

(a) the solvent (b) water trapped in the paper
(c) the capillary tube (d) air

Answer: (b) water trapped in the paper — Water combined with the cellulose.

13. The mobile phase in paper chromatography is:

(a) the solvent moving up the paper (b) water in the paper
(c) cellulose (d) the pencil line

Answer: (a) the solvent moving up the paper — It carries the solutes.

14. The spot of sample is placed on the:

(a) top edge (b) solvent front
(c) baseline (d) back of the paper

Answer: (c) baseline — Near the bottom of the paper.

15. The sample is spotted on the paper with a:

(a) capillary tube (b) burette
(c) beaker (d) thermometer

Answer: (a) capillary tube — A small spot is needed.

16. At the start, the solvent level must be:

(a) at the top of the paper (b) above the spot
(c) below the spot of the sample (d) exactly on the spot

Answer: (c) below the spot of the sample — So that the sample is not washed into the solvent.

17. Components which dissolve better in the water of the paper:

(a) change colour (b) travel the farthest
(c) evaporate (d) do not travel very far

Answer: (d) do not travel very far — They stay in the stationary phase.

18. Components more soluble in the solvent:

(a) stay on the baseline (b) move further up the paper
(c) sink in the solvent (d) react with cellulose

Answer: (b) move further up the paper — They move with the mobile phase.

19. Colourless substances are made visible by a:

(a) locating agent (b) carrier gas
(c) detector (d) capillary tube

Answer: (a) locating agent — It gives colour to them.

20. R_f stands for:

(a) relative formula (b) rate of flow
(c) reaction frequency (d) retardation (retention) factor

Answer: (d) retardation (retention) factor — Given in the book.

21. The R_f value is the distance travelled by the substance divided by the:

(a) time taken (b) length of the paper
(c) distance travelled by the solvent (d) mass of the sample

Answer: (c) distance travelled by the solvent — Definition.

22. The R_f value has:

(a) the unit cm (b) no unit
(c) the unit s (d) the unit mol

Answer: (b) no unit — It is a ratio.

23. Under the same solvent and temperature, the R_f value of a substance:

(a) does not change (b) doubles each time
(c) becomes zero (d) is unpredictable

Answer: (a) does not change — So it identifies the substance.

24. A spot moves 2 cm and the solvent 8 cm. The R_f value is:

(a) 4.0 (b) 0.25
(c) 0.40 (d) 16

Answer: (b) 0.25 — 2 ÷ 8.

25. The R_f value of a substance is always:

(a) negative (b) greater than 1
(c) exactly 10 (d) less than or equal to 1

Answer: (d) less than or equal to 1 — The spot cannot travel further than the solvent.

26. The highest point reached by the solvent is the:

(a) retention time (b) baseline
(c) solvent front (d) peak

Answer: (c) solvent front — It is marked before drying.

27. Paper chromatography can separate:

(a) only metals
(b) only coloured substances
(c) only gases
(d) both coloured and colourless substances

Answer: (d) both coloured and colourless substances — Colourless ones need a locating agent.

28. The paper used in chromatography is made of:

(a) silica (b) cellulose
(c) alumina (d) glass

Answer: (b) cellulose — It holds water.

28.3 THIN-LAYER CHROMATOGRAPHY

29. The stationary phase in TLC is:

(a) benzene (b) water in paper
(c) helium (d) alumina or silica

Answer: (d) alumina or silica — A white solid layer.

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30. The formula of alumina is:

(a) $AlCl_3$
(b) SiO_2
(c) Al_2O_3
(d) $Al(OH)_3$

Answer: (c) Al_2O_3 — Aluminium oxide.

31. The formula of silica is:

(a) SiO_2
(b) Al_2O_3
(c) $SiCl_4$
(d) Na_2SiO_3

Answer: (a) SiO_2 — Silicon dioxide.

32. In TLC, the slurry is spread on a:

(a) filter paper
(b) glass slide
(c) metal column
(d) plastic tube

Answer: (b) glass slide — Then dried in an oven.

33. Compared with paper chromatography, TLC is:

(a) the same speed
(b) slower
(c) faster
(d) impossible at room temperature

Answer: (c) faster — Stated in the book.

34. The baseline on a TLC plate is drawn with:

(a) ink pen
(b) pencil and ruler
(c) marker
(d) ninhydrin

Answer: (b) pencil and ruler — Pencil does not dissolve in the solvent.

35. The solvent for TLC is placed in a:

(a) detector
(b) burette
(c) injector port
(d) chromatography tank

Answer: (d) chromatography tank — The slide is dipped in it.

36. In TLC, polar substances are:

(a) adsorbed more strongly
(b) not adsorbed
(c) carried fastest
(d) evaporated

Answer: (a) adsorbed more strongly — By the polar stationary phase.

37. In TLC, non-polar substances:

(a) react with silica
(b) stay at the baseline
(c) move further up the slide
(d) turn purple

Answer: (c) move further up the slide — They are more soluble in the non-polar solvent.

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38. The principle of TLC is:

(a) partition between gas and liquid (b) adsorption and solubility
(c) ion exchange (d) sedimentation

Answer: (b) adsorption and solubility — Given in the comparison table.

39. Ninhydrin gives a purple colour with:

(a) halogens (b) alkanes
(c) metals (d) amino acids

Answer: (d) amino acids — A locating agent.

40. The colour given by ninhydrin with amino acids is:

(a) purple (b) yellow
(c) green (d) brown

Answer: (a) purple — Stated in the book.

41. Some colourless spots turn brown with:

(a) helium (b) ninhydrin
(c) water (d) iodine vapours

Answer: (d) iodine vapours — A locating method.

42. Some colourless substances glow under:

(a) infrared light (b) ultraviolet light
(c) red light (d) radio waves

Answer: (b) ultraviolet light — A locating method.

43. Components on a TLC plate are identified by comparing their:

(a) densities
(b) masses
(c) R_f values with standard values
(d) boiling points

Answer: (c) R_f values with standard values — Stated in the book.

44. TLC can check the progress of a reaction by showing whether:

(a) reactants are consumed or products are formed
(b) the temperature rises
(c) the pressure falls
(d) the colour of the flask changes

Answer: (a) reactants are consumed or products are formed — An application.

45. In forensic science, TLC is used to identify:

(a) fingerprints only (b) drugs and post-blast debris
(c) the age of bones (d) voice samples

Answer: (b) drugs and post-blast debris — Also blood, sweat and semen.

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46. In the food industry, TLC detects:

(a) preservatives, colourants and sweeteners
(b) calories
(c) temperature
(d) weight

Answer: (a) preservatives, colourants and sweeteners — An application.

47. TLC is mainly a:

(a) spectroscopic technique
(b) quantitative technique
(c) preparative technique
(d) qualitative technique

Answer: (d) qualitative technique — It gives only an idea of the amount.

48. TLC cannot clearly separate compounds with:

(a) different polarities
(b) different colours
(c) close R_f values
(d) very different R_f values

Answer: (c) close R_f values — Resolution is not always sharp.

49. TLC is not suitable for large-scale separation because:

(a) it works only for gases
(b) it is too fast
(c) it needs a carrier gas
(d) only small amounts can be applied

Answer: (d) only small amounts can be applied — A limitation.

50. The reproducibility of TLC is reduced by:

(a) the colour of the tank
(b) humidity and temperature
(c) the length of the ruler
(d) the brand of pencil

Answer: (b) humidity and temperature — And cleanliness of the environment.

51. More than one sample on the same TLC plate:

(a) can be spotted at different places
(b) cannot be run
(c) must be mixed first
(d) need separate tanks

Answer: (a) can be spotted at different places — Stated in the book.

52. With alumina and benzene, which travels further: butanone or pentane?

(a) both equally
(b) butanone
(c) pentane
(d) neither moves

Answer: (c) pentane — Pentane is non-polar; butanone is adsorbed strongly.

53. After the run, the TLC slide is:

(a) washed with water
(b) dried with a dryer
(c) heated to redness
(d) dipped in acid

Answer: (b) dried with a dryer — After marking the solvent front.

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54. Using a more polar solvent, the R_f values of polar amino acids:

(a) increase (b) decrease
(c) become zero (d) stay exactly the same

Answer: (a) increase — They are more soluble in the mobile phase.

28.4 GAS CHROMATOGRAPHY

55. Gas chromatography is used to separate:

(a) metals (b) proteins
(c) polymers (d) gases and volatile liquids

Answer: (d) gases and volatile liquids — And solids in gaseous form.

56. The mobile phase in GC is:

(a) an inert carrier gas (b) a polar solvent
(c) water (d) silica

Answer: (a) an inert carrier gas — Such as helium or nitrogen.

57. A suitable carrier gas for GC is:

(a) oxygen (b) helium
(c) chlorine (d) ammonia

Answer: (b) helium — It is inert.

58. The stationary phase in GC is a:

(a) solid metal
(b) volatile polar liquid
(c) non-volatile, non-polar hydrocarbon liquid
(d) carrier gas

Answer: (c) non-volatile, non-polar hydrocarbon liquid — Bonded to silica or alumina particles.

59. The stationary liquid in GC must have a:

(a) high conductivity (b) low boiling point
(c) strong colour (d) high boiling point

Answer: (d) high boiling point — So that it remains liquid in the hot column.

60. In GC the sample is introduced through the:

(a) recorder (b) detector
(c) injector port (d) exit of the column

Answer: (c) injector port — It is vaporised there.

61. The sample in GC is vaporised by:

(a) an oven (b) a magnet
(c) a lamp (d) a pump

Answer: (a) an oven — At the injector port.

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62. The GC column is:

(a) short and straight (b) long and coiled
(c) a flat plate (d) a sheet of paper

Answer: (b) long and coiled — Packed with tiny silica particles.

63. The particles in the GC column have a:

(a) bright colour (b) very small surface area
(c) magnetic field (d) large surface area

Answer: (d) large surface area — Separation occurs over it.

64. In GC, substances that interact more strongly with the stationary phase:

(a) move more slowly (b) move faster
(c) do not move at all (d) decompose

Answer: (a) move more slowly — They have longer retention times.

65. The separated substances in GC are recorded by the:

(a) injector (b) detector
(c) oven (d) carrier gas

Answer: (b) detector — As peaks on a chromatogram.

66. The graph obtained in GC is called a:

(a) histogram (b) spectrum
(c) chromatogram (d) thermogram

Answer: (c) chromatogram — It shows peaks.

67. The time taken by a substance to pass through the column is the:

(a) peak area (b) R_f value
(c) half-life (d) retention time

Answer: (d) retention time — Definition.

68. The proportion of a component in GC is given by its:

(a) peak area (b) retention time
(c) colour (d) boiling point

Answer: (a) peak area — GC is a quantitative technique.

69. The area of a GC peak is calculated as:

(a) 2 × base × height (b) base × height
(c) ½ × base × height (d) base ÷ height

Answer: (c) ½ × base × height — Peaks are similar to triangles.

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70. Peak height is used instead of area when peaks:

(a) are very broad
(b) are very narrow or have similar base widths
(c) overlap completely
(d) are coloured

Answer: (b) are very narrow or have similar base widths — Stated in the book.

71. Peak areas of A, B and C are 10, 30 and 60. The percentage of B is:

(a) 33 %
(b) 10 %
(c) 60 %
(d) 30 %

Answer: (d) 30 % — (30 ÷ 100) × 100.

72. GC is used to check competing athletes for:

(a) haemoglobin
(b) steroids
(c) calcium
(d) vitamins

Answer: (b) steroids — Stated in the book.

73. BAC tested by GC stands for:

(a) blood alcohol content
(b) blood acid concentration
(c) basic amino compound
(d) bacterial count

Answer: (a) blood alcohol content — A forensic application.

74. Petrochemical plants use GC to analyse:

(a) the mass of tankers
(b) the colour of pipes
(c) the composition of fuels
(d) soil texture

Answer: (c) the composition of fuels — An industrial application.

75. In the food industry, GC detects spoilage by identifying:

(a) the price
(b) the colour of packets
(c) the mass of food
(d) metabolic by-products of bacteria or fungi

Answer: (d) metabolic by-products of bacteria or fungi — Before they are visible.

76. In clinical settings, GC analyses compounds in a patient's:

(a) breath or urine
(b) hair colour
(c) fingerprints
(d) voice

Answer: (a) breath or urine — To diagnose metabolic disorders.

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77. GC works only for substances that are:

(a) ionic solids (b) volatile
(c) polymers (d) insoluble

Answer: (b) volatile — A limitation.

78. Samples for GC must be:

(a) magnetic (b) coloured
(c) thermally stable (d) radioactive

Answer: (c) thermally stable — They must not decompose at high temperature.

79. Proteins cannot be analysed by GC because they:

(a) do not vaporise easily (b) are too small
(c) are gases (d) are inert

Answer: (a) do not vaporise easily — Large biomolecules.

80. A limitation of GC is its:

(a) slow speed
(b) low sensitivity
(c) high instrument and maintenance cost
(d) poor resolution

Answer: (c) high instrument and maintenance cost — Stated in the book.

81. Nian Bin was proven innocent with the help of:

(a) distillation (b) paper chromatography
(c) titration (d) gas chromatography

Answer: (d) gas chromatography — The food sample showed no poison.

82. The rat poison in the Nian Bin case was a:

(a) cyanide (b) fluoroacetate
(c) chloride (d) sulfate

Answer: (b) fluoroacetate — Stated in the book.

83. Nian Bin was released in:

(a) 1994 (b) 2005
(c) 2013 (d) 2020

Answer: (c) 2013 — Eight years after the accusation in 2005.

84. Mixtures of gases can be separated by GC at:

(a) 1000 °C only (b) room temperature
(c) absolute zero (d) the boiling point of silica

Answer: (b) room temperature — Stated in the book.

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28.5 COMPARISON OF THE THREE TYPES

85. The principle of paper chromatography is:

(a) ion exchange (b) adsorption only
(c) partition (d) precipitation

Answer: (c) partition — From the comparison table.

86. The principle of gas chromatography is:

(a) gas–liquid partition (b) adsorption + solubility
(c) filtration (d) sublimation

Answer: (a) gas–liquid partition — From the comparison table.

87. The fastest of the three techniques is:

(a) all are equal (b) paper chromatography
(c) thin layer chromatography (d) gas chromatography

Answer: (d) gas chromatography — Very fast.

88. The slowest of the three techniques is:

(a) thin layer chromatography (b) paper chromatography
(c) gas chromatography (d) all are equal

Answer: (b) paper chromatography — From the comparison table.

89. The mobile phase is a liquid in:

(a) none of them (b) gas chromatography only
(c) all three (d) paper chromatography and TLC

Answer: (d) paper chromatography and TLC — In GC it is a gas.

90. The stationary phase is a non-volatile liquid in:

(a) gas chromatography (b) paper chromatography
(c) thin layer chromatography (d) none of them

Answer: (a) gas chromatography — From the comparison table.

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